

TrES-3b

R-2392

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_120329 · created 2026-08-20T23:44:36+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456015.8798 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.145 +/- 0.024
Transit depth (Rp/Rs)^2	2.11 +/- 0.7 [%]
Orbital Inclination (inc)	83.01 +/- 1.37
Airmass coefficient 1 (a1)	1.017 +/- 0.013
Airmass coefficient 2 (a2)	-0.0087 +/- 0.0062
Scatter in the residuals of the lightcurve fit is	1.29 %
Best Comparison Star	#2 - [60, 246]
Optimal Aperture	3.59
Optimal Annulus	5.97
Transit Duration (day)	0.06 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.145 $\pm$ 0.024 vs 0.16309 (deviation 0.75 σ)
Duration fit vs published	0.06 d vs 0.0567 d (deviation 0.28 σ)
Mid-time O–C	1.7 min
Depth SNR	6.0
Residual scatter	1.29 %

Light curve

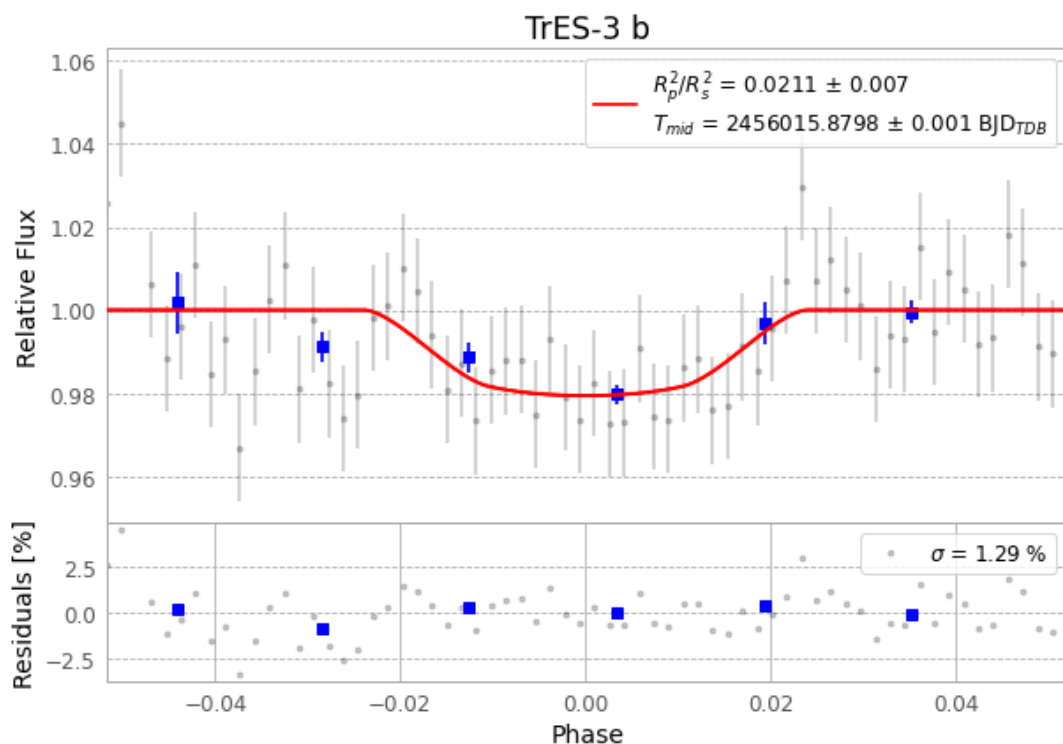


Figure 1 — FinalLightCurve_TrES-3 b_2012-03-29.png (SHA-256 5efa9e14ac0aded7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [258, 154], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1e26559cbde5f20f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

66 FITS frames (43 MB), TRES-3120329072712.FITS → TRES-3120329104212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-120329.log (f745ace26338...), FinalLightCurve_TrES-3 b_2012-03-29.png (5efa9e14ac0a...), FinalLightCurve_TrES-3 b_2012-03-29.csv (53e09f60b24b...)

Compute resources

Wall time 143.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-20 23:44Z.